higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 4

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This marking guideline consists of 6 pages.

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QUESTION 1: ROAD DRAINAGE

1.1 1.1.1 Erode

1.1.2 Debris

1.1.3 Fords

1.1.4 Kerb

1.1.5 Spanish windlass

1.1.6 Geotextile

1.1.7 Sand blankets

1.1.8 Manholes

1.1.9 Oversaturated

1.1.10 Shoulder

(10 × 1) (10)

1.2
- Reduced tyre grip on road surface.
- Higher possibility of aqua planning.
- Surface degradation.
- Formation of potholes.

(Any 3 × 1) (3)

1.3 At a slope steeper than the road surface. (1)

1.4 1.4.1
- To provide easy access into the drain when maintenance must be done.
- To provide easy access into the drain to clear or unblock the drain.
- For inspection purpose into the drain.

(Any 2 × 1) (2)

1.4.2
- Circular chamber
- Square chamber
- Rectangular chamber

(Any 2 × 1) (2)

1.4.3
- Frame
- Cover (lid)
- Steps
- Reducer slab
- Adaptor slab
- Chamber section
- Shaft section
- Starter ring
- Spacer slab

(Any 3 × 1) (3)

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1.5 1.5.1 Pipe underdrains.
Consist of perforated pipe placed at the bottom of a narrow trench√ and backfilled with coarse material.√ (2)

1.5.2 Drilled drains.
Consist of perforated metal pipes placed in holes cut into fill or slopes after construction. (2)

1.5.3 French drains.
Consist of trenches backfilled with aggregate, very coarse sand or gravel. (2)

1.6
- Rock particles
- Water
- Air (3)

[30]

QUESTION 2: ROAD PAVEMENT

2.1
- Where heavy or concentrated wheel loads are carried by the pavement layers.
- Where high frequency of turning or slewing movement is expected.
- Where volume of traffic is high.
- Where ready access to underground services is required.
- Where subgrade conditions are poor.
- Where the pavement must withstand severe in-service conditions.
- Where the appearance and aesthetic qualities of the pavement play an important part in the design consideration. (Any 5 × 1) (5)

2.2
- Used to fill the gaps or open spaces between the pacing blocks.
- Help to resist stresses between the blocks when under load or stress. (2)

2.3 2.3.1 45 - 60 MPa

2.3.2 30 - 45 MPa

2.3.3 45 - 60 MPa

2.3.4 25 - 30 MPa

2.3.5 25 - 30 MPa

(5 × 1) (5)

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2.4
- Must have a thickness of less than 20 mm not more than 30 mm.
- Must not plastic.
- Must be free from any substances that may damage course to the block.
- Must be loose without any pre-compaction and any pre-compaction must be removed and replaced.
- Must be maintained at moisture content 4% and 8%.
- Evenly laid. (Any 5 × 1) (5)

2.5.1 Bitumen.
Is a viscous or solid dark brown substance✓ obtained from the distillation of very specific quality crude oil.✓ (2)

2.5.2 Tar.
Is made from the destructive combustion ✓ of coal or wood.✓ (2)

2.6
- Ductility test
- Viscosity test
- Flashpoint test
- Solubility test
- Penetration test
- Softening point test. (Any 4 × 1) (4)

2.7
- Pen-graded bitumen
- Bitumen emulsion
- Modified bitumen
- Cut-back bitumen (Any 3 × 1) (3)

2.8
- Excellent resistance of water
- Non - scuffing under normal conditions.
- Superior weathering ability
- Flexile over a wide temperature range
- Repaired with relative ease
- Produces a duration surface finish for all weather conditions (6)

2.9 60% bitumen emulsion (1)

2.10
- Expansion joints
- Crack control joints
- Construction joints
- Contraction joints (4)

2.11
- Mobile asphalt plant
- Permanent mixing plant (Any 1 × 1) (1)

[40]

QUESTION 3: ROAD MAINTENANCE AND REHABILITATION

3.1
- Surface failure
- Potholes (2)

3.2
- Increases visibility around curves and intersections
- Increases drainage of shoulders and side drains
- Controls invader plants
- Minimises fire hazards
- Minimises security risk / hazard. (Any 2 × 1) (2)

3.3
- Slope stabilisation
- Catch pits
- Kerbs
- Culverts
- Cross fall (Any 3 × 1) (3)

3.4
- Lane closures.
- Temporary diversions (2)

3.5 3.5.1 Rehabilitation.
Is the removal of a section of a road.
The restoration of the original slope and natural drainage patterns. (2)

3.5.2 Reconstruction.
Involves the complete replacement of the entire road structure.
Applying a new pavement layer to the road. (2)

3.6
- Slippery surface
- Cracking
- Water penetration (Any 1 × 1) (1)

3.7
- Sweep or wash the surface to remove all dust and loose particles for good bonding between layers.
- Apply a single or double seal as specified. (2)

3.8
- Thin composite white topping
- Classic composite topping (2)

3.9 3.9.1 Repaving.
Consists of the heating, scarification and/or grinding of the surface to remove the top 25 – 50mm of existing asphalt.
A rejuvenated agent is added to improve the binder viscosity. (2)

3.9.2 Remixing.
Consists of the heating, scarification and/or grinding of the surface to remove the top 25 – 50mm of existing asphalt.
New virgin aggregate or new HMA is added to the recycled

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material. (2)

3.9.3 Heater scarification.
A plant heats up the pavement, scarifies the surface using a bank of non-rotating teeth.
A rejuvenated agent and mixers are added to the recycled mix. (2)

3.10 3.10.1 Traverse cracking.
Caused by:

- Improper slab lengths
- Deficient slab thickness
- Moisture and temperature variations
- Concrete with high thermal expansion
- Concrete with insufficient base support (Any 2 × 1) (2)

3.10.2 Shattered slabs and corner breaks.
Caused by:

- Concrete with insufficient slab thickness
- Concrete with insufficient base support (2)

3.10.3 Punch - outs
Caused by:

- Higher than allowable traffic loads
- Insufficient structural strength
- Deficient slab thickness
- Deficient sub-base support (Any 2 × 1) (2)

[30]

TOTAL: 100

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